

Emily Maxwell Outland

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Interests & Inspirations

I am driven by a simple conviction: that humans and autonomous systems are better together than either is alone. My research sits at the intersection of human factors and autonomy — asking how we design machines that genuinely serve the humans who work alongside them, rather than the other way around.

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|--------------------------------|--|
| Space & Exploration | Space analogs, long-duration mission human factors, human-machine teams in operational environments |
| Explainable Autonomy | Interpretable AI for non-expert users, transparency in autonomous decision-making, trust calibration |
| Broader Curiosities | Science communication, human-centered autonomy, physiological monitoring in extreme environments |

Education

BS in Computer Engineering

ROSE-HULMAN INSTITUTE OF TECHNOLOGY

Minors: Mathematics, Music

Terre Haute, IN

May 2025

MS in Aerospace Engineering Science

UNIVERSITY OF COLORADO BOULDER

Focus Area: Bioastronautics

Boulder, CO

Expected Graduation: May 2027

Experience

Air Force Research Lab - Space Warfare Directorate

SMART SCHOLAR

2023 - Current

- **Summer 2023:** explored explainable machine learning methods to detect anomalies in satellite telemetry data; research published
- **Summer 2024:** explored the viability of applying genetic algorithms to control system optimization; research concluded, not published
- **Summer 2025:** explored using Large Language Models (LLMs) to assist satellite operators in task identification and execution; research in progress
- **Summer 2026:** explored applications for human-machine teaming in satellite ops to best target AFRL needs; research in progress

ARIA Systems and Arquilla Group Labs

GRADUATE RESEARCHER

August 2025 - Current

Research into explainable multi-agent motion planning and teachable machine learning

Publication

An Explainable Machine Learning Approach for Anomaly Detection in Satellite Telemetry Data

E MAXWELL*, S KRICHEFF*, C PLAKS*, M SIMON

Published: 13 May 2024

- IEEE Aerospace Conference Proceedings 2024, Presented: 7 Mar 2024
- Applying explainability methods (SHAP, LIME, and Layer-wise Relevance Propagation) to novel IF-CBLOF (Isolation Forest Clustering-Based Local Outlier Factor) model and LSTM Neural Network to detect, score, and explain anomalies in satellite telemetry data

* indicates equal contribution as first authors

Projects

The Mad HATter

EXPLORING THE EFFECTS OF AUTONOMOUS AGENT PERSONALITY ON HUMAN-AGENT TEAM (HAT)

Aug 2025 – Dec 2025

DYNAMICS IN A COLLABORATIVE SETTING

- Studied how AI apology tone and error rate affect human trust, workload, and agent perception in a within-subjects human participant study
- Found apology quality significantly predicted trust; rude apologies produced *lower* trust than no apology, and higher apology frequency decreased trust independent of tone
- Results suggest good apologies can partially recover trust under high AI error, with direct implications for personality design in human-machine teaming

Explainable Sampling-Based Motion Planning

COMPARING SALIENCY AND TEXT-BASED EXPLANATIONS FOR RRT PATH UNDERSTANDING

Aug 2025 – Dec 2025

- Developed two explanation types — visual heatmaps and interactive contrastive text — to help humans understand and predict robot motion planning decisions
- Evaluated explanation effectiveness on human path prediction accuracy via a counterbalanced within-subjects study ($N = 24$) across three environment types
- Identified limitations of standard path similarity metrics in complex environments, motivating topology-aware evaluation as future work

In-Progress

Evaluating Comprehension and Explainability for Multi-Agent Explainable Path Planning on a Grid World

RESEARCH PROPOSAL EXPERIMENTAL PILOT

Current

- Design an experiment to evaluate comprehension and intuitiveness of a multi-agent explainable planner which uses path segmentation
- Incorporate validated surveys for comprehension and trust with counterbalanced explanation conditions
- Run a small pilot test to write a proposal for the continuation of this research area

Causal Reinforcement Learning with Human Expert Critique

MASTER'S THESIS

Current

- Build a Structural Causal Model (SCM) for an unmanned spaceflight relevant scenario and use Causal Reinforcement Learning (CRL) to develop a policy
- Develop an interface to translate causal policies into natural language (NL) for human interpretability
- Use human expert critique in NL to modify the SCM to update the CRL's knowledge of the scenario

Research Methods & Expertise

Experimental Design	Within-/between-subjects, Latin square counterbalancing, IRB protocols, power analysis
Statistical Analysis	Repeated-measures ANOVA, Tukey HSD, ANCOVA, Shapiro-Wilk, Levene's test
Human Factors & HRI	NASA-TLX, trust measurement (Likert composites) and calibration, human-agent teaming
Programming	Python, Julia, R, C/C++, MATLAB, Java, AI-assisted programming
ML & Motion Planning	Reinforcement learning (MDP/POMDP), sampling planners, explainable planning, multi-agent planning
Autonomous Systems	Explainable AI (XAI), LLMs for operator assistance, satellite telemetry analysis, robotics
Technical Writing	Conference paper and presentation, peer review experience, $\LaTeX$
Space Medicine	Autonomic nervous system physiology, HRV as a biomarker, psychosocial isolation in ICE environments
Languages	Native English, Elementary Japanese

Research Visions

- *How can we design AI systems for operational environments rather than demanding adaptation from the human?*
- *Can physiological signals and natural language feedback inform adaptive autonomy?*
- *What does it mean for autonomy to be truly explainable to a non-expert, and do better explanations actually produce better human-robot collaboration?*